Cybersecurity - Undergraduate Level Assessment 9 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. In a ____________________ attack, the extra data that holds some specific instructions in the memory for actions is projected by a cyber-criminal or penetration tester to crack the system.
   (a) Phishing
   (b) MiTM
   (c) Buffer-overflow
   (d) Clickjacking
2. Which of the following is not an example of presentation layer issues?
   (a) Poor handling of unexpected input can lead to the execution of arbitrary instructions
   (b) Unintentional or ill-directed use of superficially supplied input
   (c) Cryptographic flaws in the system may get exploited to evade privacy
   (d) Weak or non-existent authentication mechanisms
3. The stack is memory for storing
   (a) Local variables
   (b) Program code
   (c) Dynamically linked libraries
   (d) Global variables
4. When a wireless user authenticates to any AP, both of them go in the course of four-step authentication progression which is called ____________________
   (a) AP-handshaking
   (b) 4-way handshake
   (c) 4-way connection
   (d) wireless handshaking
5. What are the types of scanning?
   (a) Port, network, and services
   (b) Network, vulnerability, and port
   (c) Passive, active, and interactive
   (d) Server, client, and network

True / False (10 marks)

Answer True or False

6. True or False: A packet filter firewall operates primarily at the network or transport layer of the OSI model.
7. True or False: The digest created by a hash function is typically known as a Message Authentication Control (MAC).
8. True or False: A Base Signal Station is primarily used for managing data traffic rather than providing signal coverage in mobile networks.
9. True or False: Message confidentiality ensures that only the intended sender and receiver can access the content of a communication.
10. True or False: To verify a digital signature, the sender's public key is required to authenticate the signature's integrity.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to calculate the difference between the squared sum of first n natural numbers and the sum of squared first n natural numbers.
12. Write a function to check whether the given string is ending with only alphanumeric characters or not using regex.

End of Paper